

Multi-Family Deal Evaluator

Autonomous Decision Support for 2–4 Unit Residential Real Estate Investments

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Capstone Checkpoint 7.1 — Final Capstone Project Presentation | September 2026

Repository: github.com/jelanigb/carnegie-mellon-agentic-ai (MIT License)

[UNIT_03]TYPE: 1B/1B
EST. RENT: \$2,450/mo

✓ DATA MATCHED

[UNIT_02]TYPE: 2B/1.5B
EST. RENT: PENDING...

⚠ SPARSE LOCAL COMPS

[UNIT_01]TYPE: 2B/2B
EST. RENT: \$3,100/mo

✓ DATA MATCHED

Agenda & Presentation Structure

A complete walkthrough from problem motivation to architecture, live demo, and empirical results

01

Problem & Users

The small multi-family data gap, evaluation judgment challenges, and three stakeholder personas.

02

Scope & 2 Axes

System boundaries (2–4 units) and separating Software Confidence from Property Investment Merit.

03

Live Demo

Streamlit demo showing clean runs, real data degradation, human pauses, and independent axes.

04

Under the Hood

7-agent LangGraph pipeline, Rule-Gated AI, read-only MCP, and Transparent Degradation.

05

Evaluation & Next

30-case empirical evaluation harness, key accuracy findings, repository walkthrough, and next steps.

Why Multi-Family Deal Evaluation Needs Agentic AI

Small multi-family (2–4 unit) properties fall into an institutional data gap between single-family and commercial assets

01

Severe Data Parity Gap

- The 2–4 unit segment has far less available data than single-family homes or large commercial complexes.
- Sub-location disparity is extreme: while Brooklyn has dense listings, Staten Island holds only 6 listings in the entire borough.
- Administrative schedules (like HUD Fair Market Rents) lag market changes by 18+ percentage points, creating built-in valuation bias.

02

Complex Evaluation Judgment

- Evaluating a deal requires multiple chained judgment calls: comp selection, rent estimation, and forward pricing forecasts.
- No single model suffices: rents depend on unit-level amenities, while appreciation depends on macroeconomic interest rates.
- Evidence is often incomplete or conflicting across MLS sources, requiring thoughtful reconciliation.

03

Brittle Manual Tooling

- Investors traditionally rely on static spreadsheets or crude rules of thumb (e.g., the 1% rule or flat cap rates).
- Manual tools fail to account for sequential decision chains where early gaps (e.g., relaxed comp radius) change downstream validity.
- Spreadsheets discard uncertainty: once numbers are entered, all context regarding data reliability disappears.

TARGET PERSONAS

Three Personas, One Shared Report

A single unified report artifact serves three roles, with intelligent routing when issues arise



END CUSTOMER

The Investor

Primary Consumer: Reads the final generated report to make capital allocation and investment decisions.

Workflow: Clean, high-confidence reports flow directly to the Investor unedited.

Decision Focus: Answers 'Is this property a good investment?' with validated numbers and disclosures.



HUMAN REVIEWER

The Real Estate Agent

Client Advisor: Presents findings to the Investor and contextualizes property risks.

Workflow: System routes deals here when deal-level flags fire (e.g. sparse comps, wide rent bounds).

Decision Focus: Decides how to frame findings and makes final call whether to send report.



PLATFORM SUPPORT

The IT Specialist

Technical Operator: Maintains pipeline stability, external API connections, and model health.

Workflow: System routes runs here when infrastructure exceptions fire (e.g. geocoder outage).

Decision Focus: Resolves pipeline failures and unblocks report generation.

Key Architectural Detail: There is ONE report, not three. Disclosures dynamically route stopped deals to the exact desk equipped to resolve them.

SCOPE & BOUNDARIES

System Scope & Boundaries

Precise functional boundaries establish system capabilities while eliminating operational risk

IN-SCOPE CAPABILITIES

- ✓ **Asset Class:** Small multi-family residential properties (strictly 2 to 4 units).
- ✓ **Primary Metros:** Chicago, Los Angeles, Cleveland, and New York City.
- ✓ **Input Format:** Ingests unstructured text listing descriptions (MLS, broker notes, flyers).
- ✓ **Output Deliverable:** Investor-grade report covering property profile, comps, valuation, forward forecast, and transparent disclosures.
- ✓ **Out-of-Market Ingestion:** Accepts listings from other US markets with explicit out-of-market disclosure flags.

EXPLICIT BOUNDARIES & GUARDRAILS

- ✗ **Commercial Excluded:** Properties > 4 units are commercial assets and are strictly rejected at pre-flight.
- ✗ **Zero Write Access:** The tool execution layer is implemented as a strictly read-only MCP server.
- ✗ **No Financial Execution:** The agent cannot transact, submit offers, or make purchases on the user's behalf.
- ✗ **Confined Risk Surface:** The entire operational risk surface is strictly confined to the text of the generated report.
- ✗ **Transparent Reporting:** The primary failure mode is ungrounded claims; mitigate via enumerated flags and confidence scoring.

Design Principle: High analytical autonomy with zero operational liability. The system advises; humans decide and transact.

CORE ARCHITECTURAL PRINCIPLE

Two Questions Kept Deliberately Apart

Separating software confidence from real estate investment quality prevents misleading conclusions

Axis 1: Software Confidence

“Can the system stand behind its own numbers?”

- **Scope:** A statement about the SOFTWARE and evidence quality.
- **Logic:** Computed from accumulated disclosure flags (warns: -0.15, criticals: -0.40).
- **Threshold:** Confidence < 0.60 or ANY critical flag pauses for Human Review.
- **Outcomes:** Reports cleanly vs. Escalates to Human Review.

Axis 2: Property Deal Quality

“Is this property a good deal?”

- **Scope:** A statement about the REAL ESTATE ASSET.
- **Logic:** Computed by deterministic rule comparing asking price against ZIP recorded sales.
- **Boundary:** NEVER reads the confidence score or escalation decision.
- **Outcomes:** Proceed | Proceed with Caution | Do Not Proceed | No Recommendation.

Case in Point: The Staten Island Deal

Staten Island escalates to human review at 0.00 confidence (Axis 1) because the corpus holds 0 comps in that borough. Yet on Axis 2, it asks 17% below its neighborhood median, yielding an unequivocal 'Proceed'.

If the two axes were blurred, a reader would mistake an infrastructure data gap for a bad investment opportunity—or conversely, buy an overpriced property because the analysis was confident. Keeping them apart is essential.

Streamlit.io Demo: 3 Key Operating Points

Navigating the interactive demo to observe clean runs, real-world data degradation, and independent axes

CLEAN BASELINE

Los Angeles (Echo Park)

- **Operating Point:** Ideal condition, high comp density.
- **Confidence:** 1.00 · Proceed
- **Disclosures:** 4 informational disclosures (rent anchor ratio, index series).
- **Demonstrates:** Clean 7-agent sequential pipeline execution with 8 matching comps.

HUMAN ESCALATION PAUSE

Staten Island (Tottenville)

- **Operating Point:** Real-world geographic comp sparsity.
- **Confidence:** 0.00 · Proceed 
- **Disclosures:** Critical sparse_comps flag after 4 progressive search relaxation iterations.
- **Demonstrates:** Real LangGraph interrupt(), routing to Real Estate Agent, and human note resumption.

INDEPENDENT AXES

Overpriced (Chicago Uptown)

- **Operating Point:** Deliberately mispriced target listing.
- **Confidence:** 1.00 · Proceed with Caution 
- **Disclosures:** Clean comp retrieval and rent model, but asks 55% over ZIP 60640 median sales.
- **Demonstrates:** Spotless software confidence alongside cautionary real estate investment verdict.

Under the Hood: Stack, Models & Data Infrastructure

A production-grade Python architecture built with LangGraph, scikit-learn, and OpenRouter

Core Stack

- **LangGraph:** StateGraph with 8 nodes, conditional edges, and SQLite checkpointing.
- **Pydantic v2:** Strictly typed state object; validation error text feeds LLM retry prompts.
- **MCP Server:** Local Model Context Protocol server exposing 4 strictly read-only tools.
- **Observability:** LangSmith integration tracing latency, spans, and tokens across every node.
- **Demo Surface:** Streamlit local web app replaying deterministic committed recordings.

Models Employed

- **Primary LLM:** nvidia/nemotron-3-nano-30b-a3b via OpenRouter (selected empirically over 7 candidates).
- **Embeddings:** sentence-transformers/all-MiniLM-L6-v2 (local cosine similarity in ChromaDB).
- **Valuation Model:** Custom scikit-learn GradientBoostingRegressor trained on 5-fold CV.
- **Forecast Evaluator:** LLM ranks enumerated historical framings; never invents raw rates.

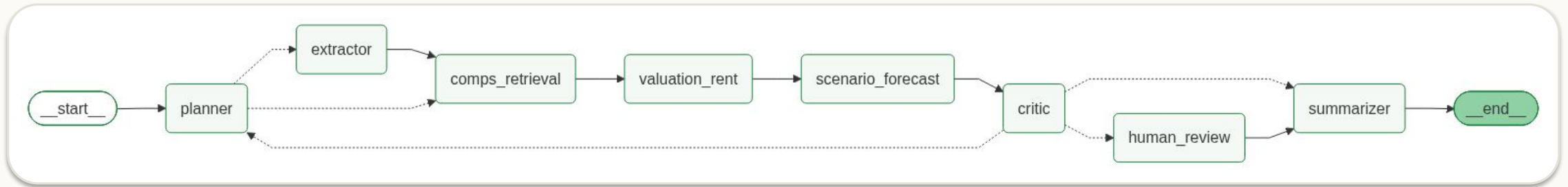
Real-World Data

- **Rent Series:** Zillow Observed Rent Index (ZORI) for ZIP market rent + HUD FMR for bedroom steps.
- **Price Benchmarks:** Redfin multi-family sales + NYC and Cook County assessor recorded deeds.
- **Geocoding:** US Census Bureau Geocoder API + Census TIGER shapefile point-in-polygon joins.
- **Training Corpus:** 99,000-listing UCI apartment-rental dataset with unit-level features.

PIPELINE TOPOLOGY

Seven-Agent Sequential State Graph

Agents execute strictly sequentially over shared Pydantic state without direct communication



01. Planner

Pre-flight deal inspector; determines required steps and sets graph routing plan.

02. Extractor

Parses unstructured listing into typed schema, geocodes address, and logs assumptions.

03. Retrieval

Adaptive hybrid RAG; progressively relaxes search criteria when sparse, logging concessions.

04. Valuation

Custom GBR rent model; predicts rent-to-market-index ratio, cross-checks against comps.

05. Forecast

Tree-of-Thought beam search over rent-growth/price scenarios; LLM ranks and prunes hypotheses.

06. Critic

Consistency checks, confidence scoring, buy/pass decisions, and routing (rework or review).

07. Summarizer

Compiles final investor report, surfacing all upstream disclosures with complete transparency.

System Invariant

Agents never invoke each other directly; all mutations occur strictly via partial state updates.

Architectural Principles & Multi-Tier Guardrails

Structural enforcement guarantees disclosure propagation, rule-gated AI, and reliable human oversight

Transparent Degradation

- **Core Philosophy:** An agent proceeding on relaxed evidence must raise a named, severity-graded disclosure flag.
- **Structural Enforcement:** LangGraph's append-only reducer makes flag loss impossible.
- **Closed Enum:** 30 typed disclosures across info (0.00), warn (-0.15), and critical (-0.40) tiers.
- **Auditability:** Upstream concessions propagate into the final report with 100% auditability.

Rule-Gated AI

- **Deterministic Primacy:** Every model call sits upstream of or alongside a deterministic rule; LLM never has the final word.
- **Schema Validation:** Extractor outputs are strictly validated against Pydantic models with up to 3 automated retries.
- **Hypothesis Ranking:** LLM ranks and prunes candidates but cannot generate arbitrary growth figures.
- **Pure Functions:** Buy/pass recommendations and scoring are pure, repeatable functions.

Guardrails & Oversight

- **Read-Only MCP:** External tool surface has zero write permissions or transaction capabilities.
- **Bounded Execution:** Graph rework loops and retries are bounded by explicit counters to prevent infinite cycles.
- **Graceful Degradation:** Live API calls fall back gracefully with typed disclosures rather than crashing.
- **Human Interrupt:** Graph halts if confidence < 0.60 or any critical flag fires, routing to Agent or IT.

Safety Strategy: Multi-layered defense. Structurally append-only disclosures, bounded graph loops, and targeted human intervention.

EMPIRICAL VALIDATION

Evaluation Harness & Quantitative Results

30-case evaluation batch on compiled production graph with pre-declared target outcomes

20 / 23

VERDICT AGREEMENT

Matched pre-declared outcome on 87% of scored cases; all 3 edge disagreements triaged and documented.

30 / 30

DISCLOSURE COVERAGE

100% coverage of all 30 typed disclosure flags in the system; zero uncovered, zero unreachable.

\$452 / mo

RENT MODEL ACCURACY

5-fold cross-validated MAE vs. \$590 baseline (23.3% improvement). Chicago \$343 vs NY \$855.

8 / 8 vs 0/8

RETRIEVAL GROUNDEDNESS

With RAG active, 8 of 8 cited comps exist in reality; with RAG disabled, 0 of 8 ungrounded comps exist.

PARAMETER SENSITIVITY & SEARCH VALUE FINDINGS

- Parameter Robustness: Grid sensitivity sweep of 160 configurations showed 63 decide all 21 cases identically. The escalation threshold can move between 0.30 and 0.70 without a single verdict changing.
- Rule Independence: Critical weight is behaviorally inert across its whole range (including zero) because critical flags escalate via an independent rule regardless of score.
- Tree-of-Thought Search Value: Across four target properties, beam search kept the default 1st-enumerated framing 0 times. On Cleveland, search adjusted the forecast base case by -22.0% on rent and +26.4% on price.

Live Execution & Engineering Production Process

End-to-end uncached model invocation, LangSmith telemetry, and human-in-the-loop AI software engineering

LIVE EXECUTION & OBSERVABILITY

- **Uncached Live Call:** Live execution of a previously unseen listing, running end-to-end.
- **Streaming Progression:** Real-time observation of sequential agent state mutations as each completes its work.
- **LangSmith Observability:** Complete distributed trace capture for every node invocation.
- **Production Verification:** Proves the system functions dynamically without relying on pre-baked caches.

7-WEEK PRODUCTION VELOCITY

- **Scope & Velocity:** Delivered 14,894 lines of Python across 84 files with 165 Git commits in a 7-week cycle.
- **Role Specialization:** Author served as Chief Architect and Data Scientist, defining pipelines, schemas, and evals.
- **Agentic Pair Programming:** Delegated code implementation execution to Anthropic Claude Code under human review.
- **Research Acceleration:** Used Google Gemini and Anthropic Claude for data strategy research and cross-checking.
- **Quality Assurance:** 107 hermetic unit tests and automated regression checking across all modules.

Engineering Insight: Strategic delegation to coding agents enabled senior engineering management oversight at 10x production velocity.

Project Summary, Limitations & Public Repository

Open-source codebase, acknowledged architectural trade-offs, and future improvements

Core Takeaway

- **Honest AI Engineering:** Building an agent that discloses its own uncertainty is the same discipline as building it honestly.
- **Empirical Falsification:** Measurement challenged three initial premises: metro selection, HUD anchor drift (+52% vs +33%), and correlation direction.
- **Structural Integrity:** Append-only reducers and rule gating ensure LLMs assist analysis without risking hallucinations.

Limitations & Next Steps

- **Data Vintage:** Current comp corpus is 2018–19; next step is integrating live MLS/aggregator feeds.
- **California Price Records:** Assessor records lack sale prices under Prop 13; requires commercial deed integration.
- **MCP Capabilities:** Expose evaluation harness itself as an external MCP tool for third-party agents.
- **Financial Metrics:** Ingest operating expense records to compute true Net Operating Income (NOI) and cap rates.

Public GitHub Repository

- **Repository URL:** <https://github.com/jelanigb/carnegie-mellon-agentic-ai>
- **License:** Open-Source MIT License.
- **Documentation:** Comprehensive README with architecture diagrams, setup, and evidence pointers.
- **Artifacts Included:** Complete 7-agent code, 30-case evaluation harness, 107 hermetic tests, and 3 sample markdown reports.

Special Thanks: Professor Farimani and the Carnegie Mellon University instructional team. Onwards & upwards!